Angela Qian

qian220@purdue.edu | in/angelazqian | angelazqian

EDUCATION

Purdue University | *B.S. in Computer Science, Minor in Mathematics* **Aug 2023 – May 2027**

EXPERIENCE

Undergraduate Research Assistant | *Advisor: Joseph Campbell* **Feb 2025 - Present**
- Worked on team of 7 to design an LLM-based agent with Theory of Mind to play the social deduction game Avalon
- Conducted participant studies with 45+ volunteers and applied statistical analysis to evaluate model performance
- Developed and evaluated methods for hallucination detection with 95% agreement with human annotations
- Drafted and co-authored a paper in submission to a top-tier machine learning conference

Software Engineering Intern | *Cepton Technologies Inc.* **May 2025 - Aug 2025**
- Built in Rust a real-time object tracking pipeline using LiDAR-generated point cloud data on a per-frame basis
- Developed a 2D rectangle selection tool for a 3D point cloud viewer, enabling users to select points from any camera perspective and display their coordinates dynamically by transforming 3D point data using projection matrices
- Performed real-time foreground-background segmentation masking from depth and reflectivity maps by engineering a computer vision model based on U-Net segmentation

Teaching Assistant | *Purdue University Department of Computer Science* **Jan 2025 - Present**
- CS 182: Discrete Mathematics, CS 211: Competitive Programming
- Led weekly recitation sessions, teaching problem-solving techniques and reinforcing theory taught in class
- Graded assignments for 800+ students, providing detailed feedback to improve students' understanding of material
- Held weekly office hours to clarify course material and guide students through homework challenges
- Conducted academic dishonesty investigations and interviewed students in coordination with course staff

Undergraduate Researcher | *Purdue University Vertically Integrated Projects* **Aug 2024 – Dec 2024**
- Detected image blur with custom program using convolution, the Laplacian filter, and the Sobel filter
- Generated clear, focused photos from blurred images with algorithms based on Fourier Transforms
- Presented in Purdue's 2024 Fall Undergraduate Research Exposition

Competition Mathematics Instructor | *AlphaStar Academy* **July 2024 – Aug 2024**
- Delivered lectures to 50+ students on Algebra, Number Theory, Geometry, and Counting/Probability
- Held review sessions with practice problems to prepare students for the AMC10, AMC12, and AIME exams
- Presented feedback to students and parents, highlighting focus areas for continued academic growth

PUBLICATIONS

Bayesian Social Deduction with Graph-Informed Language Models
Shahab Rahimirad, Guven Gergerli, Lucia Romero, **Angela Qian**, Matthew Lyle Olson, Simon Stepputtis, Joseph Campbell (2025) *Under Review*

PROJECTS

2048 AI Player | *PyTorch, Jupyter Notebook, JavaScript* **June 2025 - Present**
- Developed a neural network to play 2048 using imitation learning trained on a custom dataset of my own gameplay
- Built model in PyTorch to predict actions from board states, and exported it via ONNX to deploy in TensorFlow.js
- Currently extending the project with reinforcement learning to fine-tune decision making through self-play

ScribbleScore Handwriting Improvement | *Python, OpenCV, Numpy, React, Flask* **Feb 2025**
- Worked in a group of 4 over the course of a 36-hour hackathon (Boilermake XII)
- Designed bounding-box logic to group words and measure spacing consistency across clusters
- Implemented Sobel edge detection, line of best fit analysis to assess writing straightness, extracting text from images, and computing score

SKILLS

Languages: English (Native), Mandarin Chinese (Native)
Programming: Python, Rust, C/C++, JavaScript, Typescript, x86_64 Assembly, SQL, MongoDB, HTML, CSS, Java
Developer Tools: Git, GitHub, Jupyter Notebook, Linux, Overleaf, React, LaTeX, VIM, FFmpeg, MySQL, Neo4J, Next.js, TailwindCSS, VS Code, Eclipse, Agile & Scrum
Libraries: Matplotlib, OpenCV, PyTorch, TorchVision, Pandas, NumPy